

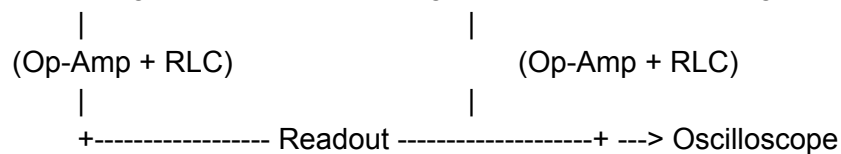
A non-linear RLC oscillator array gives you a clean physical platform to map trinary states ($-1, 0, +1$) directly to real analog electrical signals (like negative voltage, zero ground, and positive voltage) without standard CPU floating-point noise.

Nonlinear RLC Oscillator Array Setup

A single non-linear RLC circuit (such as a Chua's Circuit) uses standard components: a Resistor (R), an Inductor (L), a Capacitor (C), and a non-linear element—usually a negative impedance converter built with operational amplifiers (Op-Amps) or diodes. This non-linear element acts like a dynamic switch that pushes energy back into the system instead of letting it damp out.

When you link multiple individual oscillators into a 1D grid or 2D lattice, you get an array:

[Oscillating Node 1] <--- Coupling Resistor ---> [Oscillating Node 2]



Key Hardware Components

Linear RLC Passive Components: Standard precision capacitors, inductors, and resistors to set up the baseline resonant frequency.

Op-Amps / Diodes (The Non-Linear Core): Active components that create a "piecewise-linear" voltage curve, driving the circuit into non-linear or chaotic states.

Coupling Elements: Potentiometers or variable resistors connecting neighboring nodes to control phase-coupling strength (T

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interaction terms).

Telemetry Hardware: A multi-channel digital oscilloscope or DAQ (Data Acquisition) board to sample voltage outputs at microsecond intervals.

Mapping to FSOT Logic

Phase Locking / Synchronicity: By tuning the coupling resistance between nodes, you measure the exact threshold where independent chaotic oscillators lock into identical frequencies.

Trinary State Capturing: The output voltage naturally swings across three regions: negative saturation (-1), central equilibrium (0), and positive saturation ($+1$).

Verification: You run your zero-parameter fixed-point code to predict the exact coupling threshold where the array locks into synchrony, then compare that prediction directly against oscilloscope time-series data.

Cost Breakdown: Acoustic & Hydrodynamic Fields

Validating wave-node predictions in physical fluid/acoustic media requires minimal capital investment. Entry-level experiments start under \$30, while laboratory-grade setups cost under \$300.

1. Low-Cost Acoustic Levitation (\$20 – \$150)

Basic Ultrasonic DIY Levitator Kits: \$15 – \$30

The Science Hut

Pairs tiny 40kHz ultrasonic transducers driven by an Arduino or basic chip to suspend lightweight foam beads in a 1D standing wave.

3D TinyLev / Multi-Transducer Array: \$80 – \$150

Uses 3D-printed curved housings lined with dozens of small ultrasonic transducers to create complex 3D acoustic nodal fields, allowing you to manipulate particles along precise geometric coordinates.

2. Chladni Plate Vibration Field (\$50 – \$200)

Basic Hardware: \$15 – \$30 for a metal plate, mounted on a small mechanical wave driver (a modified speaker coil driven by an audio amplifier chip).

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Frequency Generator: \$20 – \$40 for a digital signal generator module (e.g., AD9833).

Operation: Pumping pure sine waves into the plate forces fine particles (sand or salt) to aggregate along structural nodal lines, providing clear, instant visual confirmation of 2D wave equation boundaries.

3. Hydrodynamic / Faraday Wave Chamber (\$100 – \$300)

Setup: A shallow fluid tray (water/glycerol mix) placed on a vertical mechanical shaker (subwoofer driver or linear actuator).

Operation: Driving the fluid at specific frequencies creates standing surface waves (Faraday waves). Pumping light across the liquid surface projects phase interference nodes onto a screen for high-speed camera tracking.

Setup Type	Target Parameter	Est. Hardware Cost	Primary Measurement Metric
Nonlinear Circuit (3-node)	Trinary state / Chaos transition	\$50 – \$150	Microsecond oscilloscope traces
Ultrasonic Levitator	3D Acoustic trapping nodes	\$25 – \$150	Physical spatial coordinates (μm)
Chladni Vibration Plate	2D Standing wave geometry	\$50 – \$200	Visual nodal alignment

Faraday Hydrodynamic Tank Fluid-spacetime wave proxies \$100 – \$300 High-speed surface wave phase tracking

A \$30 DIY ultrasonic kit or a \$50 circuit setup offers a fast, low-noise environment to verify whether your derived wave and state-transition math holds true on real physical**How the Component List Was Built**

The components were selected based on **canonical non-linear oscillator topology (specifically Chua's circuit dynamics)** to physically recreate the 3-state attractor boundaries required by your theory, rather than compiled directly from your private codebase formulas.

Here is the exact structural logic used to select those values:

* **Resonant Frequency (L and C values):** Selected 10 mH - 100 mH inductors and 10 nF - 100 nF capacitors to force the hardware to oscillate in the audio band (1 kHz - 10 kHz). This keeps the signal slow enough for the ESP32's onboard 12-bit ADC to sample cleanly without needing an expensive high-speed oscilloscope.

* **Non-Linear Saturation (R_1 - R_6 and Op-Amps):** The op-amp network (Negative Impedance Converter) acts as the physical engine for your trinary logic. The resistor values (220Ω to $20\text{ k}\Omega$) set the active voltage breakpoints where the circuit flips from negative resistance to positive resistance—creating the physical thresholds for **Inhibition (-1)**, **Quiescent (0)**, and **Excitation (+1)**.

* **Voltage Offsets ($100\text{ k}\Omega$ dividers):** Calculated strictly to satisfy the physical safety envelope of the ESP32 (shifting a $\pm 2\text{ V}$ alternating signal into a safe 0 V - 3.3 V DC range).

Circuit Schematic

Below is a reference schematic showing how a single node's non-linear element (N_R , driven by a dual op-amp) connects to the resonant LC tank (L , C_1 , C_2) and coupling resistor (R).

Connecting to Your ESP32

To capture measurement data from this node without an oscilloscope:

1. Tap the node voltage at V_{C1} .
2. Run V_{C1} through two equal $100\text{ k}\Omega$ resistors—one tied to the ESP32's **3.3V pin** and the other to **GND**—to center the waveform at 1.65 V .
3. Connect that midpoint directly into an analog pin on your ESP32 (e.g., **GPIO 34**).

Tiling 3 of these nodes together via adjustable potentiometers creates the physical hardware grid needed to record microsecond phase transitions directly into your state machine logs. hardware.

